

```
In [12]: import pandas as pd

df = pd.read_csv("titanic.csv")

df.head()
```

```
Out[12]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05

```
In [8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   PassengerId      891 non-null   int64  
1   Survived         891 non-null   int64  
2   Pclass           891 non-null   int64  
3   Name             891 non-null   object  
4   Sex              891 non-null   object  
5   Age              714 non-null   float64 
6   SibSp            891 non-null   int64  
7   Parch            891 non-null   int64  
8   Ticket           891 non-null   object  
9   Fare             891 non-null   float64 
10  Cabin            204 non-null   object  
11  Embarked         889 non-null   object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [9]: df.describe()
```

Out[9]:

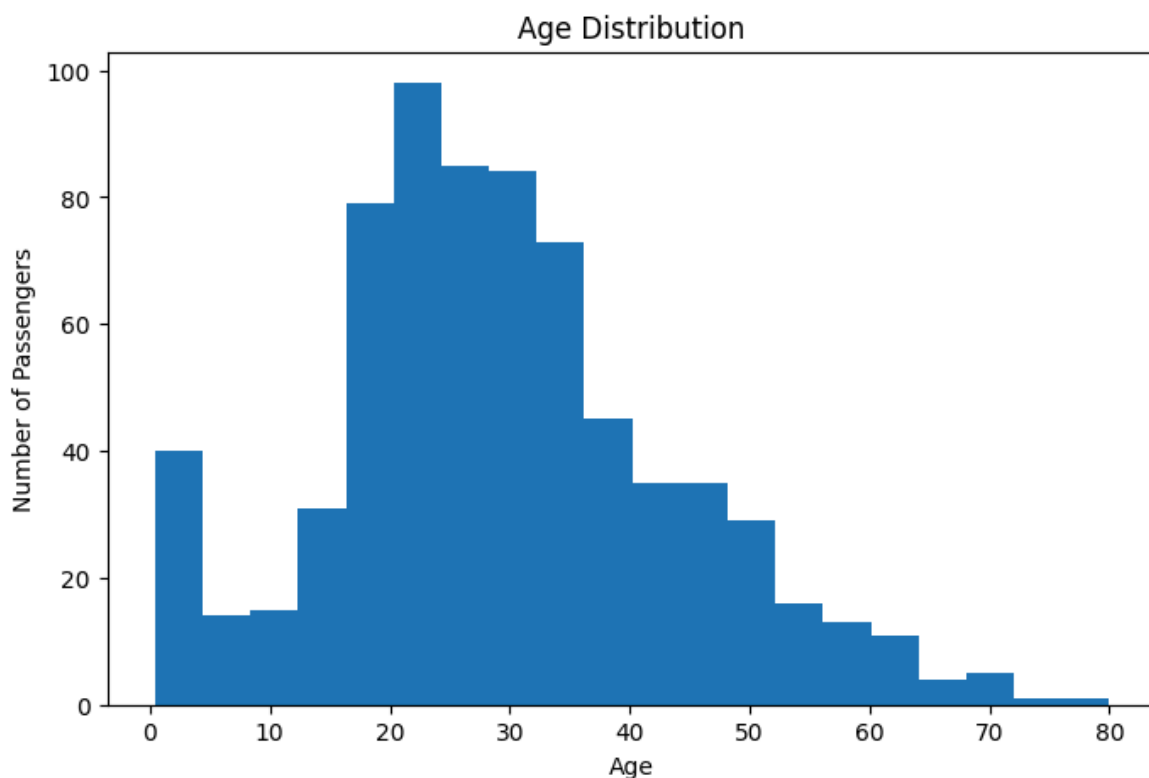
	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.200000
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.690000
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.450000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.320000

In [10]: `df["Sex"].value_counts()`

Out[10]: Sex
 male 577
 female 314
 Name: count, dtype: int64

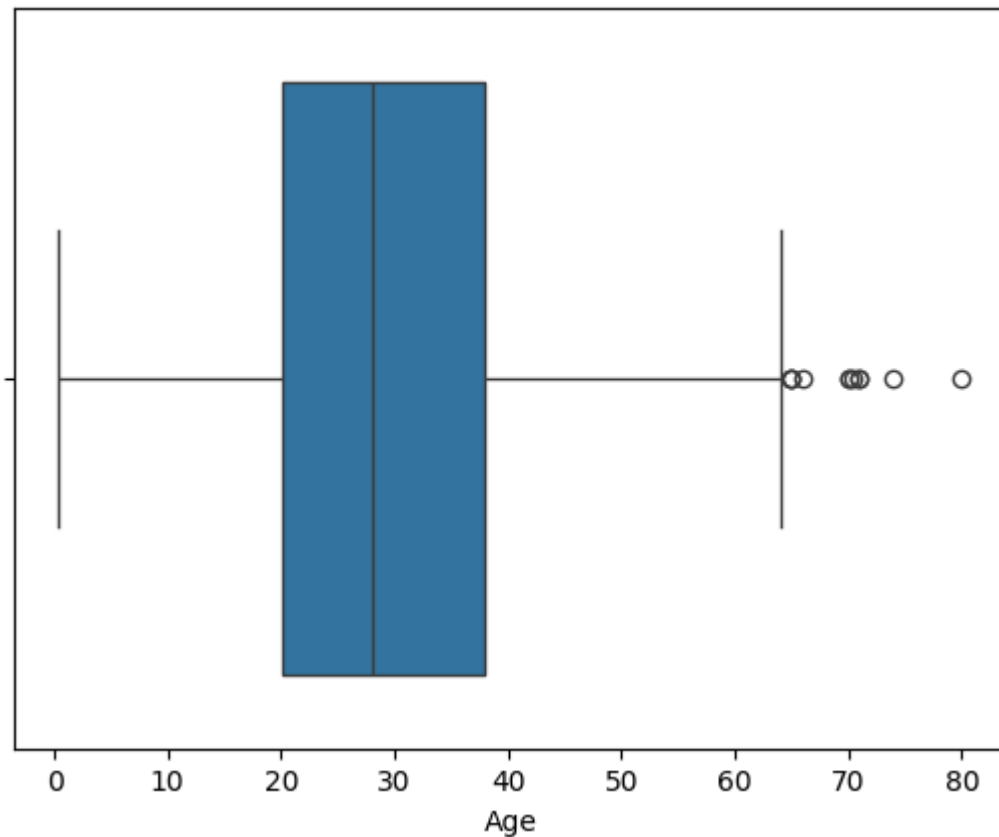
In [13]: `import matplotlib.pyplot as plt`

```
plt.figure(figsize=(8,5))
plt.hist(df["Age"], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



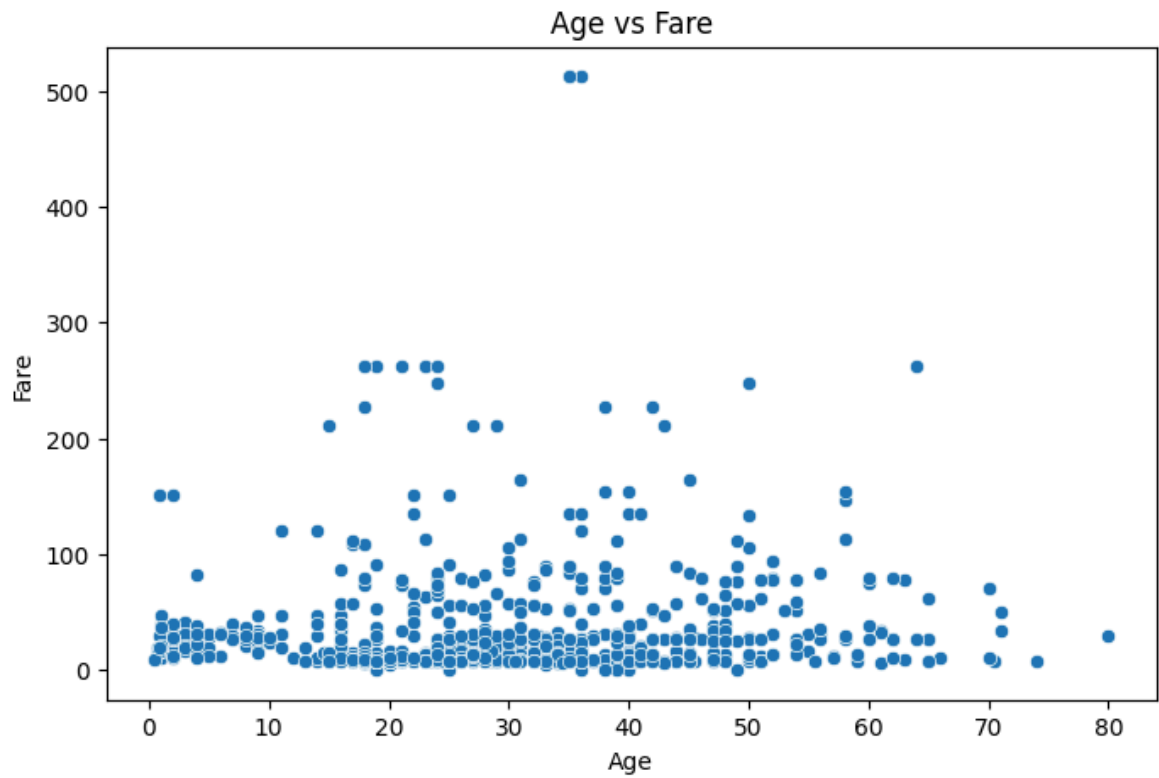
```
In [8]: import matplotlib.pyplot as plt
import seaborn as sns

sns.boxplot(x=df["Age"])
plt.show()
```



```
In [10]: import matplotlib.pyplot as plt
import seaborn as sns

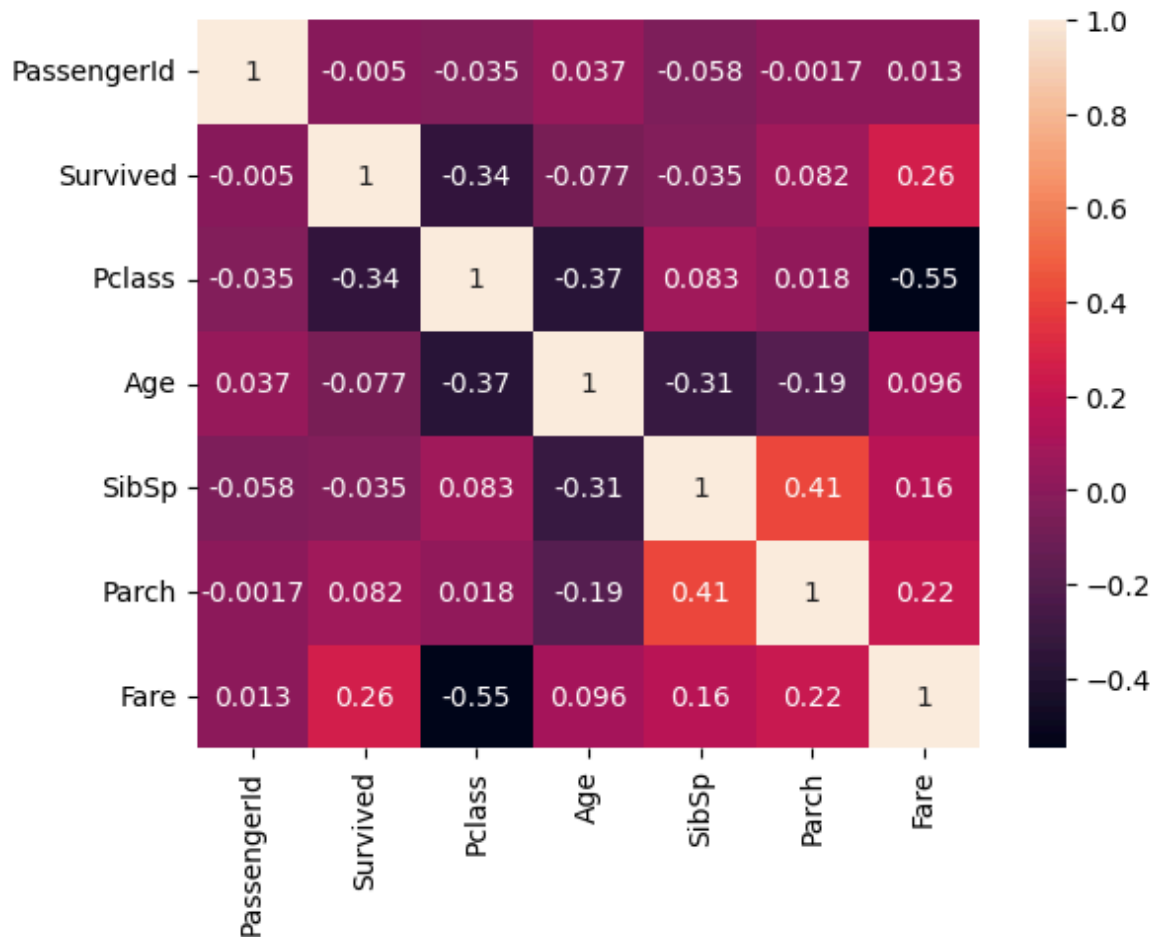
plt.figure(figsize=(8,5))
sns.scatterplot(x=df["Age"], y=df["Fare"])
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



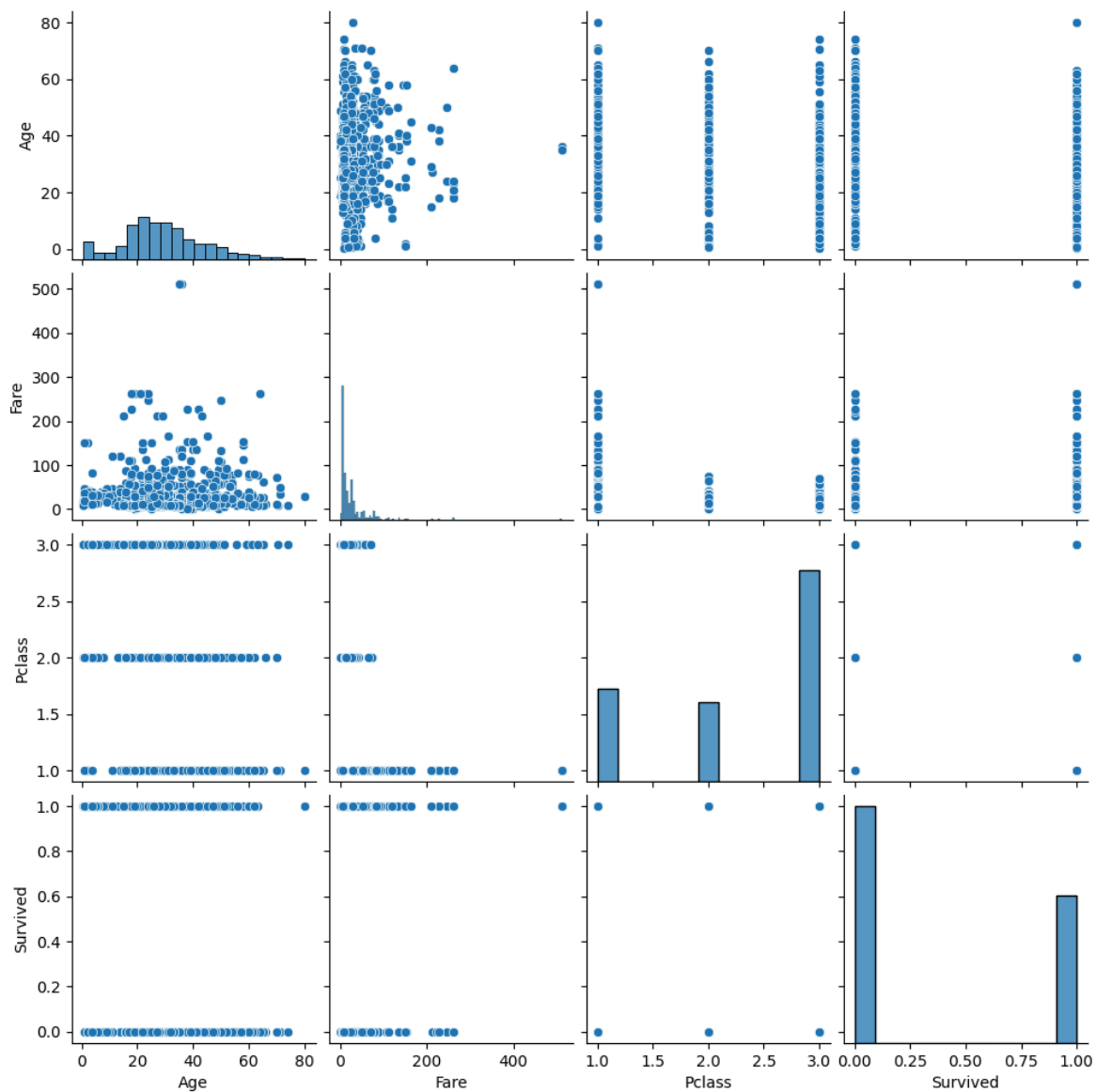
```
In [14]: correlation = df.corr(numeric_only=True)

sns.heatmap(correlation, annot=True)

plt.show()
```



```
In [12]: sns.pairplot(df[["Age", "Fare", "Pclass", "Survived"]])  
plt.show()
```



```
In [ ]:
```